



Revised

Generation Interconnection

System Impact Study Report

for

Queue Project AE2-130

ROCKPORT 765 KV

480 MW Capacity / 800 MW Energy

September, 2022

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1 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances, a generator interconnection may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection, may also contribute to the need for the same network reinforcement. Cost allocation rules for network upgrades can be found in PJM Manual 14A, Attachment B. The possibility of sharing the reinforcement costs with other projects may be identified in the feasibility study, but the actual allocation will be deferred until the impact study is performed.

The Interconnection Customer seeking to interconnect a wind generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

PJM utilizes manufacturer models to ensure the performance of turbines is properly captured during the simulations performed for stability verification and, where applicable, for compliance with low voltage ride through requirements. Turbine manufacturers provide such models to their customers. The list of manufacturer models PJM has already validated is contained in Attachment B of Manual 14G. Manufacturer models may be updated from time to time, for various reasons such as to reflect changes to the control systems or to more accurately represent the capabilities turbines and controls which are currently available in the field. Additionally, as new turbine models are developed, turbine manufacturers provide such new models which must be used in the conduct of these studies. PJM needs adequate time to evaluate the new models in order to reduce delays to the System Impact Study process timeline for the Interconnection Customer as well as other Interconnection Customers in the study group. Therefore, PJM will require that any Interconnection Customer with a new manufacturer model must supply that model to PJM, along with a \$10,000 fully refundable deposit, no later than three (3) months prior to the starting date of the System Impact Study (See Section 4.3 for starting dates) for the Interconnection Request which shall specify the use of the new model. The Interconnection Customer will be required to submit a completed dynamic model study request form (Attachment B-1 of Manual 14G) in order to document the request for the study.

2 General

The Interconnection Customer has proposed a Solar and Storage generating facility located in Spencer County, Indiana. The installed photovoltaic generation facilities will have a total capability of 800 MW with 480 MW of this output being recognized by PJM as Capacity. The project also includes 200MW of AC-connected battery storage. It is assumed that at times the Interconnection Customer will charge the batteries from the grid. The proposed in-service date for this project is 12/01/2022. This study does not imply a TO commitment to this in-service date.

The objective of this System Impact Study is to determine budgetary cost estimates and approximate construction timelines for identified transmission facilities required to connect the proposed generating facilities to the AEP transmission system. These reinforcements include the Attachment Facilities, Local Upgrades, and Network Upgrades required for maintaining the reliability of the AEP transmission system.

Queue Number	AE2-130
Project Name	ROCKPORT 765 KV
State	Indiana
County	Spencer
Transmission Owner	AEP
MFO	800
MWE	800
MWC	480
Fuel	Solar; Storage
Basecase Study Year	2022

2.1 Point of Interconnection

AE2-130 will interconnect with the AEP transmission system at the Rockport 765 kV substation.

To accommodate the interconnection at the Rockport 765 kV substation, the substation will have to be expanded requiring the installation of one 765 kV circuit breaker (see Figure 1). Installation of associated protection and control equipment, 765kV line risers, SCADA, and 765 kV revenue metering will also be required.

2.2 Cost Summary

The AE2-130 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$2,200,000
Direct Connection Network Upgrade	\$3,500,000
Non Direct Connection Network Upgrades	\$0
Allocation for New System Upgrades	\$8,144,000
Contribution for Previously Identified Upgrades	\$541,000
Total Costs	\$14,385,000

The estimates are preliminary in nature, as they were determined without the benefit of detailed engineering studies. Final estimates will require an on-site review and coordination to determine final construction requirements.

3 Transmission Owner Scope of Work

4 Attachment Facilities

The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
765kV Revenue Metering	\$500,000
Generator lead first span exiting the POI station, including the first structure outside the fence	\$1,700,000
Total Attachment Facility Costs	\$2,200,000

5 Direct Connection Cost Estimate

The total preliminary cost estimate for the Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Install one new 765 kV Circuit Breaker. Installation of associated protection and control equipment, 765kV line risers and SCADA.	\$3,500,000
Total Direct Connection Facility Costs	\$3,500,000

6 Non-Direct Connection Cost Estimate

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
None*	\$0
Total Non-Direct Connection Facility Costs	\$0

*AEP will evaluate if a portion of the scope of work at Rockport 765kV substation will be classified as Non-Direct Connection in the Facilities Study.

7 Incremental Capacity Transfer Rights (ICTRs)

Will be determined at a later study phase

8 Schedule

It is anticipated that the time between receipt of executed Agreements and Commercial Operation may range from 12 to 18 months if no line work is required. If line work is required, construction time would generally be between 24 to 36 months after Agreement execution.

9 Interconnection Customer Requirements

It is understood that the Interconnection Customer is responsible for all costs associated with this interconnection. The costs above are reimbursable to AEP. The cost of the Interconnection Customer's generating plant and the costs for the line connecting the generating plant to the transmission system are not included in this report; these are assumed to be the Interconnection Customer's responsibility.

The Generation Interconnection Agreement does not in or by itself establish a requirement for American Electric Power to provide power for consumption at the developer's facilities. A separate agreement may be reached with the local utility that provides service in the area to ensure that infrastructure is in place to meet this demand and proper metering equipment is installed. It is the responsibility of the developer to contact the local service provider to determine if a local service agreement is required.

Requirement from the PJM Open Access Transmission Tariff:

1. An Interconnection Customer entering the New Services Queue on or after October 1, 2012 with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.
2. The Interconnection Customer may be required to install and/or pay for metering as necessary to properly track real time output of the facility as well as installing metering which shall be used for billing purposes. See Section 8 of Appendix 2 to the Interconnection Service Agreement as well as Section 4 of PJM Manual 14D for additional information.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.2 AEP Requirements

The Interconnection Customer will be required to comply with all AEP Revenue Metering Requirements for Generation Interconnection Customers. The Revenue Metering Requirements may be found within the “Requirements for Connection of New Facilities or Changes to Existing Facilities Connected to the AEP Transmission System” document located at the following link:

<http://www.pjm.com/~media/planning/plan-standards/private-aep/aep-interconnection-requirements.ashx>

11 Network Impacts

The Queue Project AE2-130 was evaluated as a 800.0 MW (Capacity 480.0 MW) injection at Rockport 765 kV substation in the AEP area. Project AE2-130 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE2-130 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Load Flow

12 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

#	Type	Contingency Name	Affected Area	Facility Description	Bus		Cir.	PF	Loading %		Rating		MW Con.	FG App.
					From	To			Initial	Final	Type	MVA		
1	N-1	AEP_P1-2_#363	AEP - AEP	05DARWIN-05EUGENE 345 kV line	243216	243221	1	AC	94.75	103.49	NR	1195	104.54	
2	N-1	AEP_P1-2_#363	AEP - AEP	05SULLIVAN-05DARWIN 345 kV line	247712	243216	1	AC	94.75	103.49	NR	1195	104.54	

13 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

#	Type	Contingency Name	Affected Area	Facility Description	Bus		Cir.	PF	Loading %		Rating		MW Con.	FG App.
					From	To			Initial	Final	Type	MVA		
3	LFFB	AEP_P4_#1760_05JEFRSO 765_A	AEP - OVEC	05JEFRSO-06CLIFTY 345 kV line	242865	248000	Z1	DC	94.52	111.09	ER	2354	386.32	2
4	LFFB	AEP_P4_#8648_05JEFRSO 765_B	AEP - AEP	05DARWIN-05EUGENE 345 kV line	243216	243221	1	DC	99.62	114.23	ER	1195	174.54	4
5	LFFB	AEP_P4_#8648_05JEFRSO 765_B	AEP - AEP	05SULLIVAN-05DARWIN 345 kV line	247712	243216	1	DC	99.62	114.23	ER	1195	174.54	5

14 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

#	Type	Contingency	Affected Area	Facility Description	Bus		Cir.	PF	Loading %		Rating		MW Con.	FG App.
		Name			From	To			Initial	Final	Type	MVA		
6	LFFB	AEP_P4_#1760_05JEFRSO 765_A	OVEC - OVEC	06CLIFTY- 06DEARB1 345 kV line	248000	248001	1	AC	108.0 3	115.78	ER	956	87.22	6
7	N-1	AEP_P1-2_#363	LGEE - OVEC	7TRIMBLE CO- 06CLIFTY 345 kV line	324114	248000	1	DC	138.8 4	140.96	ER	1451	36.01	7
8	N-1	AEP_P1-2_#363	LGEE - OVEC	7TRIMBLE CO- 06CLIFTY 345 kV line	324114	248000	1	DC	138.8 4	140.96	ER	1451	36.01	
9	LFFB	AEP_P4_#3128_05EUGENE 345_A2	MISO AMIL - AEP	J1180 TAP- 05SULLIVAN 345 kV line	966790	247712	1	AC	127.2 6	128.68	ER	1466	31.58	8

15 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

#	Type	Contingency	Affected Area	Facility Description	Bus		Cir.	PF	Loading %		Rating		MW Con.	FG App.
		Name			From	To			Initial	Final	Type	MVA		
10	N-1	AEP_P1-2_#363	AEP - AEP	05DARWIN- 05EUGENE 345 kV line	243216	243221	1	DC	98.79	113.37	NR	1195	174.23	
11	N-1	AEP_P1-2_#363	AEP - AEP	05SULLIVAN- 05DARWIN 345 kV line	247712	243216	1	DC	98.79	113.37	NR	1195	174.23	

#	Type	Contingency	Affected Area	Facility Description	Bus		Cir.	PF	Loading %		Rating		MW Con.	FG App.
		Name			From	To			Initial	Final	Type	MVA		
12	N-1	AEP_P1-2_#363	AEP - MISO AMIL	05SULLIVAN-J1180 TAP 345 kV line	247712	966790	1	DC	106.1	125.99	NR	1466	291.91	
13	N-1	AEP_P1-2_#363	LGEE - OVEC	7TRIMBLE CO-06CLIFTY 345 kV line	324114	248000	1	AC	143	146.57	ER	1451	60.02	
14	N-1	AEP_P1-2_#363	MISO AMIL - CE	7BROKAW-AD2-153 TAP 345 kV line	348847	937160	1	AC	115.24	117.97	ER	1528	48.9	
15	N-1	AEP_P1-2_#363	CE - CE	AD2-153 TAP-Z2-087 TAP 345 kV line	937160	917500	1	AC	114.96	117.7	ER	1528	48.9	
16	N-1	AEP_P1-2_#286	MISO AMIL - AEP	J1180 TAP-05SULLIVAN 345 kV line	966790	247712	1	AC	119.83	121.1	NR	1466	29.55	
17	Non	Non	MISO AMIL - AEP	J1180 TAP-05SULLIVAN 345 kV line	966790	247712	1	AC	111.14	112.64	NR	1334	28.75	
18	Non	Non	MISO AMIL - AEP	J1180 TAP-05SULLIVAN 345 kV line	966790	247712	1	AC	111.14	112.64	NR	1334	28.75	

16 Steady-State Voltage Requirements

(Summary of the VAR requirements based upon the results of the steady-state voltage studies)

To be determined in the Facilities Study Phase

17 Stability and Reactive Power Requirement for Low Voltage Ride Through

(Summary of the VAR requirements based upon the results of the dynamic studies)

To be determined in the Facilities Study Phase

18 Light Load Analysis

Light Load Studies (applicable to wind, coal, nuclear, and pumped storage projects).

None.

19 System Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

Facility	Upgrade Description	Cost	Cost Allocation	Upgrade Number
05DARWIN-05EUGENE 345 kV line (from bus 243216 to bus 243221 ckt 1)	<u>AEP</u> AEP SE rating is 1335 MVA. <u>AEP Reinforcement:</u> <u>Project ID:</u> S1579 <u>Description:</u> Disconnect Darwin 345 kV station from the Eugene – Sullivan 345 kV line and retire all AEP owned equipment at Darwin. <u>Type:</u> FAC <u>Cost:</u> \$0 <u>Time Estimate:</u> 0 Months <u>Ratings:</u> 1335 MVA SE	\$0 +	\$0 +	S1579 N6628
05SULLIVAN-05DARWIN 345 kV line (from bus 247712 to bus 243216 ckt 1)	<u>AEP Reinforcement:</u> <u>Project ID:</u> N6628 <u>Description:</u> Add a 765 kV breaker 'B2' at Jefferson 765 kV to eliminate stuck breaker contingency 'AEP_P4_#8648_05JEFRSO 765_B' <u>Type:</u> FAC <u>Cost:</u> \$4,000,000 <u>Time Estimate:</u> 0 Months AE2-130 is the driver for this upgrade.	4,000,000	4,000,000	
05JEFRSO-06CLIFTY 345 kV line (from bus 242865 to bus 248000 ckt Z1)	<u>AEP End</u> AEP SE rating is 2354 MVA. <u>OVEC End</u> OVEC SE rating is 1868 MVA. <u>AEP Reinforcement:</u> <u>2021 Baseline Upgrade B2878:</u> Upgrade the Clifty Creek 345 kV risers <u>Project Type:</u> FAC <u>Cost:</u> \$ 0 <u>Time Estimate:</u> 0 Months <u>Ratings:</u> 2255 MVA SE <u>Notes:</u> 1. As changes to the PJM queue process occur (such as prior queued projects withdrawing from the queue, reducing in size, etc.) AE2-130 could receive cost allocation.	\$0 + \$2,000,000 + \$244,000 +	\$0 + \$2,000,000 + \$244,000 +	B2878 N4106.1 N4106.3 N4106.4

	3. Although Queue Project AE2-130 may not presently have cost responsibility for this upgrade, Queue Project AE2-130 may need this upgrade in-service to be deliverable to the PJM system. 4. If Queue Project AE2-130 comes into service prior to completion of the upgrade, Queue Project AE2-130 will need an interim study. <u>AEP Reinforcement:</u> Project ID: N4106.1 Description: Replace 4 Clifty switches. Type: FAC Cost: \$2,000,000 Time Estimate: 12-18 Months Ratings: 2354 MVA SE AE2-130 is the driver for this upgrade. <u>AEP Reinforcement:</u> Project ID: N4106.3 Description: A sag check will be required for the ACSR ~ 2156 ~ 84/19 ~ BLUEBIRD Conductor Section 1 to determine if the line section can be operated above its emergency rating. Past/preliminary sag study results (from Z1-070 Facilities Study): Jefferson – Clifty Creek 345 kV line work will include one location of grading to remediate clearance location of concern in span 1 to 2. Type: FAC Cost: \$244,000 Time Estimate: 6-12 Months Ratings: 2826 MVA SE AE2-130 is the driver for this upgrade. <u>AEP Reinforcement:</u> Project ID: N4106.4 Description: Rebuild the 0.75 mile ACSR ~ 2156 ~ 84/19 ~ BLUEBIRD @ 284 F - Conductor section 1. Type: FAC Cost: \$1,900,000 Time Estimate: 12-18 Months Ratings: 2826 MVA SE AE2-130 is the driver for this upgrade.	\$1,900,000	\$1,900,000	
	Total	\$8,144,000	\$8,144,000	

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

Facility	Upgrade Description	Cost	Cost Allocation	Upgrade Number
06CLIFTY-06DEARB1 345 kV line (from bus 248000 to bus 248001 ckt 1)	<u>AEP Reinforcement:</u> Project ID: B2943 Description: Perform a LIDAR study on the line. LIDAR study results show that various mitigations are required involving various structures. Type: FAC Cost: \$0 Time Estimate: ISD: 5/21/2020 Ratings: 1099 MVA SE Notes: 1. As changes to the PJM queue process occur (such as prior queued projects withdrawing from the queue, reducing in size, etc.) AE2-130 could receive cost allocation. 3. Although Queue Project AE2-130 may not presently have cost responsibility for this upgrade, Queue Project AE2-130 may need this upgrade in-service to be deliverable to the PJM system. 4. If Queue Project AE2-130 comes into service prior to completion of the upgrade, Queue Project AE2-130 will need an interim study. <u>OVEC Reinforcement:</u> Project ID: N7979 Description: Replace one 1600 A Wavetrap at Clifty Creek. Type: FAC Cost: \$200,000 Time Estimate: 12-18 Months Ratings: 1281 MVA SE AE2-130 is the driver for this upgrade.	\$0 + \$200,000	\$0 + \$200,000	B2943
7TRIMBLE CO-06CLIFTY 345 kV line (from bus 324114 to bus 248000 ckt 1)	<u>OVEC</u> No OVEC end upgrade required. The line is owned by LG&E. <u>LG&E</u> A LG&E Affected System Study will be required for AE2-130. The AE2-130 customer will need to sign onto a LG&E Affected System Study Agreement. LG&E will determine if there are any LG&E system impacts, including on the Trimble – Clifty LG&E-OVEC tie line. Final LG&E Impacts and necessary LG&E system upgrade(s) will be determined once the LG&E affected system study is completed by LG&E.	TBA	TBA	N/A
	<u>Ameren/MISO end:</u> No Ameren/MISO upgrade required. SE rating is 1466 MVA. <u>AEP Reinforcement:</u> Project ID: N5808			

J1180 TAP-055SULLIVAN 345 kV line (from bus 966790 to bus 247712 ckt 1)	Description: A Eugene stuck breaker contingency causing the overload, AEP can eliminate the Eugene stuck breaker contingency by requiring a new string, expanding the 345 kV buses, add two (2) additional 345 kV circuit breakers, and repositioning the Bunsonville 345 kV line between the two new breakers on the new string. Type: FAC Cost: \$5,000,000 Time Estimate: Ratings: The cost allocation is as follows:			\$5,000,000	\$341,000	N5808
	Queue	MW Contribution	Percentage of cost			
	AD1-100	48.6	10.48%			
	AD1-133	23.9	5.16%			
	AD2-100	37.7	8.13%			
	AD2-159	0	0.00%			
	AE1-113	20.3	4.38%			
	AE1-163	24.2	5.22%			
	AE1-172	19.5	4.21%			
	AE1-205	20	4.31%			
	J955	109.2	23.56%			
	J976	22.9	4.94%			
	J991	30.66	6.61%			
	J1115	18.41	3.97%			
	J1139	19.98	4.31%			
	J1180	36.6	7.90%			
	AE2-130	31.6	6.82%			
Total				\$5,200,000	\$541,000	

Affected Systems

20 Affected Systems

20.1 TVA

None

20.2 Duke Energy Progress

None

20.3 MISO

The October 2019 AE2 Group – Preliminary MISO Report has identified AE2-130 impacts on MISO. Please refer to the MISO October 2019 AE2 Group – Preliminary MISO Report for details.

20.4 LG&E

An LG&E affected system study will be required for AE2-130. The AE2-130 customer will be required to sign a LG&E affected system study agreement to determine if there are any LG&E impacts .

21 Contingency Descriptions

Contingency Name	Contingency Definition
AEP_P1-2_#286	CONTINGENCY 'AEP_P1-2_#286' OPEN BRANCH FROM BUS 243221 TO BUS 348885 CKT 1 / 243221 05EUGENE 345 348885 7BUNSONVILLE 345 1 END
AEP_P1-2_#363	CONTINGENCY 'AEP_P1-2_#363' OPEN BRANCH FROM BUS 243208 TO BUS 243209 CKT 1 / 243208 05JEFRSO 765 243209 05ROCKPT 765 1 END
AEP_P4_#1760_05JEFRSO 765_A	CONTINGENCY 'AEP_P4_#1760_05JEFRSO 765_A' OPEN BRANCH FROM BUS 243207 TO BUS 243208 CKT 1 / 243207 05GRNTWN 765 243208 05JEFRSO 765 1 OPEN BRANCH FROM BUS 242924 TO BUS 243208 CKT 1 / 242924 05HANG R 765 243208 05JEFRSO 765 1 END
AEP_P4_#3128_05EUGENE 345_A2	CONTINGENCY 'AEP_P4_#3128_05EUGENE 345_A2' OPEN BRANCH FROM BUS 243221 TO BUS 249504 CKT 1 / 243221 05EUGENE 345 249504 08CAYSUB 345 1 OPEN BRANCH FROM BUS 243221 TO BUS 348885 CKT 1 / 243221 05EUGENE 345 348885 7BUNSONVILLE 345 1 END
AEP_P4_#8648_05JEFRSO 765_B	CONTINGENCY 'AEP_P4_#8648_05JEFRSO 765_B' OPEN BRANCH FROM BUS 243208 TO BUS 243209 CKT 1 / 243208 05JEFRSO 765 243209 05ROCKPT 765 1 OPEN BRANCH FROM BUS 243208 TO BUS 242865 CKT 2 / 243208 05JEFRSO 765 242865 05JEFRSO 345 2 OPEN BRANCH FROM BUS 242865 TO BUS 248000 CKT Z1 / 242865 05JEFRSO 345 248000 06CLIFTY 345 Z1 END

Short Circuit

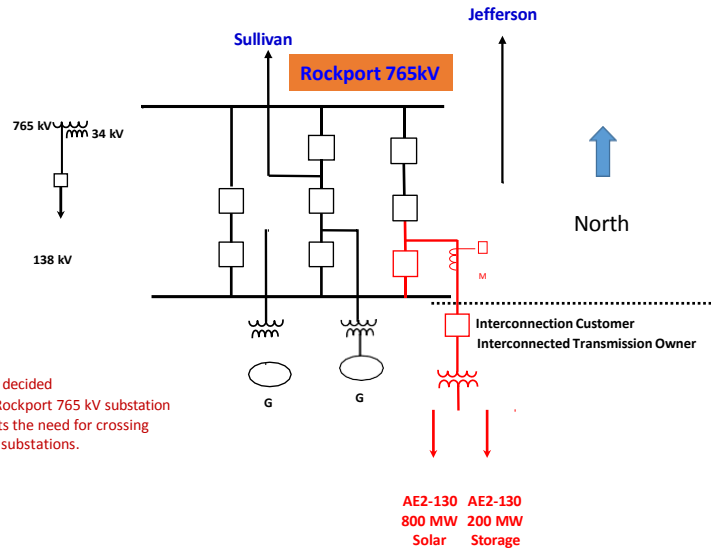
22 Short Circuit

The following Breakers are overduty

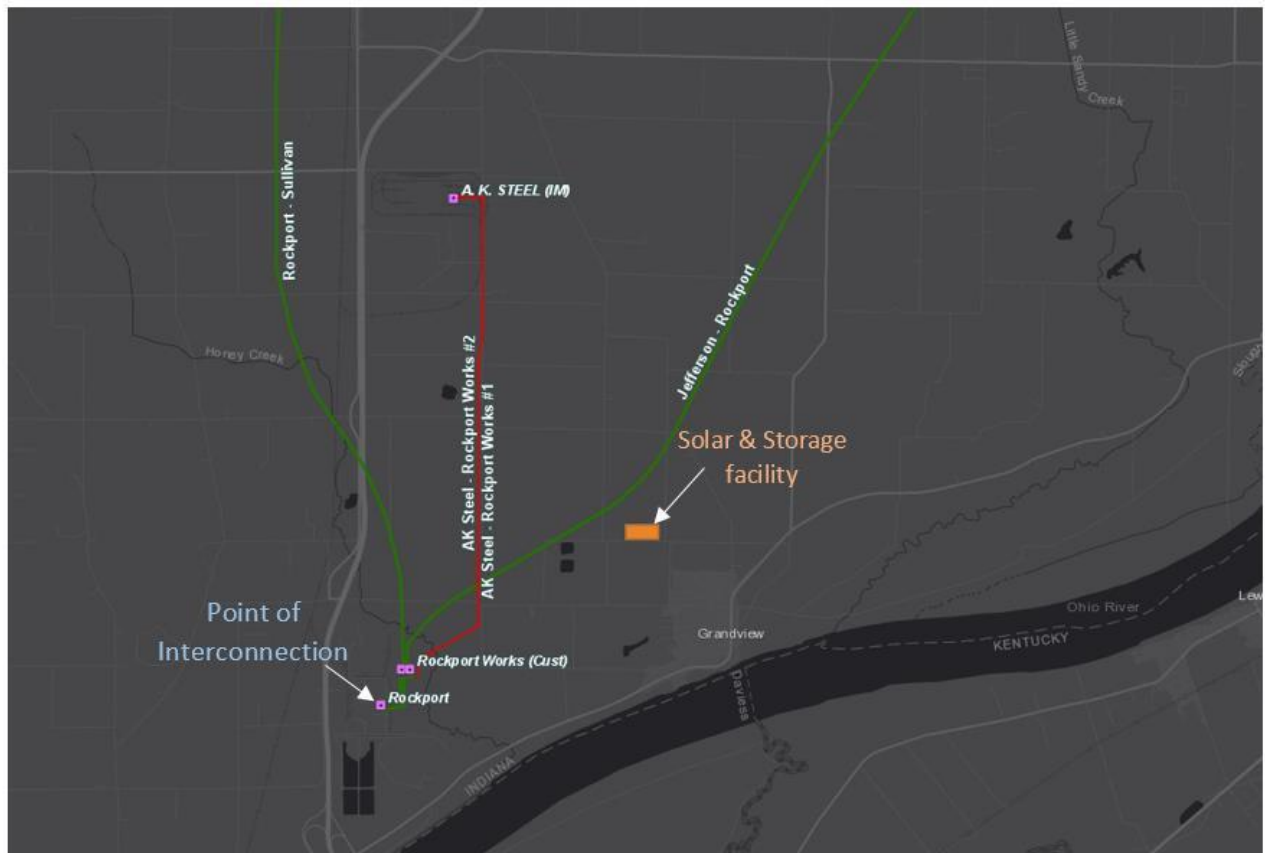
Bus Number	Bus Name	BREAKER	Type	Capacity (Amps)	Duty Percentage Post Queue	Duty Percentage Pre Queue

23 Single Line Diagram

AE2-130 Point of Interconnection Rockport 765kV Substation



24 Figure 2: Point of Interconnection Map



25 Appendices

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the Appendices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the Appendices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators.

It should be noted the project/generator MW contributions presented in the body of the report and appendices sections are full contributions, whereas the loading percentages reported in the body of the report, take into consideration the commercial probability of each project as well as the ramping impact of "Adder" contributions.

Appendix 2

(AEP - OVEC) The 05JEFRSO-06CLIFTY 345 kV line (from bus 242865 to bus 248000 ckt Z1) loads from 94.52% to 111.09% (**DC power flow**) of its emergency rating (2354 MVA) for the line fault with failed breaker contingency outage of 'AEP_P4_#1760_05JEFRSO 765_A'. This project contributes approximately 386.32 MW to the thermal violation.

CONTINGENCY 'AEP_P4_#1760_05JEFRSO 765_A'

OPEN BRANCH FROM BUS 243207 TO BUS 243208 CKT 1 / 243207 05GRNTWN 765 243208 05JEFRSO 765 1

OPEN BRANCH FROM BUS 242924 TO BUS 243208 CKT 1 / 242924 05HANG R 765 243208 05JEFRSO 765 1

END

Bus Number	Bus Name	Full Contribution
247900	05FR-11G E	6.17
247901	05FR-12G E	6.07
247902	05FR-21G E	6.49
247903	05FR-22G E	6.21
247904	05FR-3G E	12.58
247905	05FR-4G E	9.46
247906	05MDL-1G E	10.77
247907	05MDL-2G E	5.34
247912	05MDL-3G E	5.58
247913	05MDL-4G E	5.4
243442	05RKG1	101.7
243443	05RKG2	97.5
933441	AC2-157 C	17.95
933442	AC2-157 E	29.28
935141	AD1-148	3.8
936771	AD2-100 C O1	10.6
936772	AD2-100 E O1	7.07
936971	AD2-131 C O1	0.7
936972	AD2-131 E O1	3.51
939741	AE1-205 C O1	5.32
939742	AE1-205 E O1	7.35
941341	AE2-130 C	231.79
941342	AE2-130 E	154.53
941571	AE2-154 C	2.19
941572	AE2-154 E	14.67
941731	AE2-173 O1	3.17
942111	AE2-223 C	1.24
942112	AE2-223 E	8.27
942481	AE2-261 C	14.84
942482	AE2-261 E	9.89
942601	AE2-276	11.81
LTF	BLUEG	54.17
LTF	CALDERWOOD	0.27
LTF	CANNELTON	0.32
LTF	CARR	0.58

LTF	CATAWBA	0.35
LTF	CBM-W1	20.18
LTF	CBM-W2	93.52
LTF	CHEOAH	0.26
LTF	CHILHOWEE	0.09
LTF	CHOCTAW /* 35% REVERSE 4566958 4511400	< 0.01
LTF	CIN	12.06
LTF	G-007	1.67
LTF	HAMLET	0.61
LTF	IPL	8.55
966791	J1180	10.79
954681	J949	13.19
LTF	LGE-0012019	4.67
LTF	LGE-0092018	< 0.01
LTF	MEC	24.38
LTF	MECS	4.55
LTF	O-066	10.73
LTF	RENSSELAER	0.46
290261	S-027 E	11.01
290265	S-028 E	11.01
LTF	SANTEETLA	0.08
247943	T-127 E	5.4
LTF	TRIMBLE	6.66
276150	W2-048 E	1.05
905082	W4-005 E	19.76
LTF	WEC	3.31
909052	X2-022 E	14.54
913222	Y1-054 E	-1.26
915662	Y3-099 E	0.13
915672	Y3-100 E	0.13
917502	Z2-087 E	11.02
930042	AB1-006 E	11.73
930461	AB1-087	129.88
930471	AB1-088	129.88
924041	AB2-047 C O1	2.06
924042	AB2-047 E O1	13.78
924261	AB2-070 C O1	1.99
924262	AB2-070 E O1	13.32
925771	AC1-053 C	2.
925772	AC1-053 E	13.38
926841	AC1-171 C O1	0.66
926842	AC1-171 E O1	4.44

Appendix 4

(AEP - AEP) The 05DARWIN-05EUGENE 345 kV line (from bus 243216 to bus 243221 ckt 1) loads from 99.62% to 114.23% (**DC power flow**) of its emergency rating (1195 MVA) for the line fault with failed breaker contingency outage of 'AEP_P4_#8648_05JEFRSO 765_B'. This project contributes approximately 174.54 MW to the thermal violation.

CONTINGENCY 'AEP_P4_#8648_05JEFRSO 765_B'

OPEN BRANCH FROM BUS 243208 TO BUS 243209 CKT 1 / 243208 05JEFRSO 765 243209 05ROCKPT 765 1
 OPEN BRANCH FROM BUS 243208 TO BUS 242865 CKT 2 / 243208 05JEFRSO 765 242865 05JEFRSO 345 2
 OPEN BRANCH FROM BUS 242865 TO BUS 248000 CKT Z1 / 242865 05JEFRSO 345 248000 06CLIFTY 345 Z1
 END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
243442	05RKG1	45.95
243443	05RKG2	44.05
933441	AC2-157 C	16.58
933442	AC2-157 E	27.05
941341	AE2-130 C	104.72
941342	AE2-130 E	69.81
942601	AE2-276	10.91
LTF	CARR	0.11
LTF	CBM-S1	3.32
LTF	CBM-S2	0.45
LTF	CBM-W2	30.58
LTF	CPLE	0.09
LTF	EDWARDS	0.07
LTF	G-007	0.3
LTF	IPL	4.65
966791	J1180	11.13
965101	J991	10.53
LTF	LGEE	0.41
LTF	MEC	3.16
LTF	O-066	1.9
LTF	RENSSELAER	0.09
LTF	TILTON	4.54
930461	AB1-087	119.99
930471	AB1-088	119.99

Appendix 5

(AEP - AEP) The 05SULLIVAN-05DARWIN 345 kV line (from bus 247712 to bus 243216 ckt 1) loads from 99.62% to 114.23% (**DC power flow**) of its emergency rating (1195 MVA) for the line fault with failed breaker contingency outage of 'AEP_P4_#8648_05JEFRSO 765_B'. This project contributes approximately 174.54 MW to the thermal violation.

CONTINGENCY 'AEP_P4_#8648_05JEFRSO 765_B'

OPEN BRANCH FROM BUS 243208 TO BUS 243209 CKT 1 / 243208 05JEFRSO 765 243209 05ROCKPT 765 1
 OPEN BRANCH FROM BUS 243208 TO BUS 242865 CKT 2 / 243208 05JEFRSO 765 242865 05JEFRSO 345 2
 OPEN BRANCH FROM BUS 242865 TO BUS 248000 CKT Z1 / 242865 05JEFRSO 345 248000 06CLIFTY 345 Z1
 END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
243442	05RKG1	45.95
243443	05RKG2	44.05
933441	AC2-157 C	16.58
933442	AC2-157 E	27.05
941341	AE2-130 C	104.72
941342	AE2-130 E	69.81
942601	AE2-276	10.91
LTF	CARR	0.11
LTF	CBM-S1	3.32
LTF	CBM-S2	0.45
LTF	CBM-W2	30.58
LTF	CPLE	0.09
LTF	EDWARDS	0.07
LTF	G-007	0.3
LTF	IPL	4.65
966791	J1180	11.13
965101	J991	10.53
LTF	LGEE	0.41
LTF	MEC	3.16
LTF	O-066	1.9
LTF	RENSSELAER	0.09
LTF	TILTON	4.54
930461	AB1-087	119.99
930471	AB1-088	119.99

Appendix 6

(OVEC - OVEC) The 06CLIFTY-06DEARB1 345 kV line (from bus 248000 to bus 248001 ckt 1) loads from 108.03% to 115.78% (AC power flow) of its emergency rating (956 MVA) for the line fault with failed breaker contingency outage of 'AEP_P4_#1760_05JEFRSO 765_A'. This project contributes approximately 87.22 MW to the thermal violation.

CONTINGENCY 'AEP_P4_#1760_05JEFRSO 765_A'

OPEN BRANCH FROM BUS 243207 TO BUS 243208 CKT 1

/ 243207 05GRNTWN 765 243208 05JEFRSO 765 1

OPEN BRANCH FROM BUS 242924 TO BUS 243208 CKT 1

/ 242924 05HANG R 765 243208 05JEFRSO 765 1

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
932841	AC2-111 C O1	-1.71
933441	AC2-157 C	4.29
933442	AC2-157 E	7.
941341	AE2-130 C	52.33
941342	AE2-130 E	34.89
942601	AE2-276	2.82
LTF	CARR	0.05
LTF	CBM-S1	12.54
LTF	CBM-S2	1.69
LTF	CBM-W1	5.91
LTF	CBM-W2	59.42
LTF	CIN	7.
LTF	CPLE	0.51
LTF	G-007	0.09
LTF	IPL	2.29
LTF	LGEE	6.83
LTF	MEC	7.91
LTF	O-066	0.6
LTF	RENSSELAER	0.04
247543	V3-007 C	-1.47
LTF	WEC	0.65
930461	AB1-087	31.05
930471	AB1-088	31.05
926881	AC1-175 C	-4.31

Appendix 7

(LGEE - OVEC) The 7TRIMBLE CO-06CLIFTY 345 kV line (from bus 324114 to bus 248000 ckt 1) loads from 138.84% to 140.96% (**DC power flow**) of its emergency rating (1451 MVA) for the single line contingency outage of 'AEP_P1-2_#363'. This project contributes approximately 36.01 MW to the thermal violation.

CONTINGENCY 'AEP_P1-2_#363'

OPEN BRANCH FROM BUS 243208 TO BUS 243209 CKT 1
END

/ 243208 05JEFRSO 765 243209 05ROCKPT 765 1

Bus Number	Bus Name	Full Contribution
243442	05RKG1	15.8
342900	1COOPER1 G	2.66
342903	1COOPER2 G	5.16
342918	1JKCT 1G	2.11
342921	1JKCT 2G	2.11
342924	1JKCT 3G	2.11
342927	1JKCT 4G	1.4
342930	1JKCT 5G	1.39
342933	1JKCT 6G	1.4
342936	1JKCT 7G	1.4
342939	1JKCT 9G	1.43
342942	1JKCT 10G	1.43
342945	1LAUREL 1G	1.5
932551	AC2-075 C	1.18
933441	AC2-157 C	5.71
936381	AD2-048 C	4.34
936571	AD2-072 C O1	12.31
936771	AD2-100 C O1	6.57
936971	AD2-131 C O1	0.43
939131	AE1-143 C	11.67
940041	AE1-246 C O1	14.57
940831	AE2-071 C	3.69
941341	AE2-130 C	36.01
941411	AE2-138 C	18.79
941981	AE2-210 C O1	6.47
942411	AE2-254 C O1	4.89
942481	AE2-261 C	9.14
942591	AE2-275 C O1	8.4
942601	AE2-276	3.75
942891	AE2-308 C O1	14.13
943111	AE2-339 C	3.16
LTF	CARR	0.22
LTF	CBM-S1	45.95
LTF	CBM-S2	4.74
LTF	CBM-W1	9.47
LTF	CBM-W2	191.47

<i>LTF</i>	<i>CHOCTAW</i>	<i>/* 35% REVERSE 4566958 4511400</i>	<i>< 0.01</i>
<i>LTF</i>		<i>CIN</i>	<i>28.86</i>
<i>LTF</i>		<i>CPL</i>	<i>1.33</i>
<i>LTF</i>		<i>IPL</i>	<i>15.41</i>
<i>952811</i>		<i>J759</i>	<i>11.49</i>
<i>952821</i>		<i>J762</i>	<i>35.64</i>
<i>952861</i>		<i>J783 C</i>	<i>10.9</i>
<i>953611</i>		<i>J800</i>	<i>16.58</i>
<i>953931</i>		<i>J856</i>	<i>11.02</i>
<i>LTF</i>		<i>LGEE</i>	<i>27.68</i>
<i>LTF</i>		<i>MEC</i>	<i>20.17</i>
<i>LTF</i>		<i>RENSSELAER</i>	<i>0.17</i>
<i>LTF</i>		<i>WEC</i>	<i>1.48</i>
<i>930461</i>		<i>AB1-087</i>	<i>41.29</i>
<i>930471</i>		<i>AB1-088</i>	<i>41.29</i>
<i>925981</i>		<i>AC1-074 C O1</i>	<i>4.96</i>

Appendix 8

(MISO AMIL - AEP) The J1180 TAP-05SULLIVAN 345 kV line (from bus 966790 to bus 247712 ckt 1) loads from 127.26% to 128.68% (AC power flow) of its emergency rating (1466 MVA) for the line fault with failed breaker contingency outage of 'AEP_P4_#3128_05EUGENE 345_A2'. This project contributes approximately 31.58 MW to the thermal violation.

CONTINGENCY 'AEP_P4_#3128_05EUGENE 345_A2'

OPEN BRANCH FROM BUS 243221 TO BUS 249504 CKT 1

/ 243221 05EUGENE 345 249504 08CAYSUB 345 1

OPEN BRANCH FROM BUS 243221 TO BUS 348885 CKT 1

/ 243221 05EUGENE 345 348885 7BUNSONVILLE 345 1

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
932881	AC2-115 1	1.3
932891	AC2-115 2	1.3
932921	AC2-116	0.46
933911	AD1-013 C	1.
933912	AD1-013 E	1.59
933931	AD1-016 C	0.46
933932	AD1-016 E	0.76
934051	AD1-031 C O1	2.35
934052	AD1-031 E O1	3.83
934431	AD1-067 C	0.07
934432	AD1-067 E	0.3
934701	AD1-098 C O1	3.78
934702	AD1-098 E O1	2.76
934871	AD1-116 C	0.46
934872	AD1-116 E	0.75
934971	AD1-129 C	0.47
934972	AD1-129 E	0.32
935001	AD1-133 C O1	14.14
935002	AD1-133 E O1	9.42
935141	AD1-148	7.45
934101	AD2-000 1	2.83
936291	AD2-038 C O1	1.34
936292	AD2-038 E O1	9.
936511	AD2-066 C O1	4.76
936512	AD2-066 E O1	3.17
936771	AD2-100 C O1	22.65
936772	AD2-100 E O1	15.1
936791	AD2-102 C	7.93
936792	AD2-102 E	5.29
936971	AD2-131 C O1	1.49
936972	AD2-131 E O1	7.5
937001	AD2-134 C	1.38
937002	AD2-134 E	5.51
937311	AD2-172 C	1.42
937312	AD2-172 E	1.96

937531	AD2-214 C	3.25
937532	AD2-214 E	2.17
938851	AE1-113 C O1	4.45
938852	AE1-113 E O1	15.78
938861	AE1-114 C O1	2.37
938862	AE1-114 E O1	8.1
939051	AE1-134 1	0.91
939061	AE1-134 2	0.91
939321	AE1-163 C O1	3.38
939322	AE1-163 E O1	20.75
939401	AE1-172 C O1	3.42
939402	AE1-172 E O1	16.02
939741	AE1-205 C O1	8.37
939742	AE1-205 E O1	11.56
940101	AE1-252 C O1	6.86
940102	AE1-252 E O1	4.57
940501	AE2-035 C	1.42
940502	AE2-035 E	1.96
941131	AE2-107 C	3.36
941132	AE2-107 E	2.24
941343	AE2-130 BAT	31.58
941731	AE2-173 O1	4.98
942111	AE2-223 C	1.94
942112	AE2-223 E	13.01
942421	AE2-255 C O1	1.69
942422	AE2-255 E O1	5.06
942481	AE2-261 C	31.32
942482	AE2-261 E	20.88
942602	AE2-276 BAT	13.95
942651	AE2-281 C O1	0.48
942652	AE2-281 E O1	2.96
942991	AE2-321 C	4.1
942992	AE2-321 E	2.02
LTF	BLUEG	10.33
294401	BSHIL;1U E	7.06
294410	BSHIL;2U E	7.06
LTF	CARR	0.29
274890	CAYUG;1U E	10.15
274891	CAYUG;2U E	10.15
LTF	CBM-S1	13.16
LTF	CBM-S2	2.37
LTF	CBM-W1	28.29
LTF	CBM-W2	201.07
LTF	CHOCTAW /* 35% REVERSE 4566958 4511400	< 0.01
LTF	CIN	3.88
LTF	CPL	0.57
274859	EASYR;U1 E	6.82

274860	EASYR;U2 E	6.82
LTF	G-007	0.78
LTF	GIBSON	0.01
290051	GSG-6; E	5.7
965371	J1022 C	2.47
965372	J1022 E	13.39
966041	J1094	13.77
966251	J1115 C	2.92
966252	J1115 E	15.82
966421	J1139	22.45
966791	J1180	36.73
950701	J196 C	1.32
950702	J196 E	5.3
951001	J339	6.02
954761	J468 C	7.07
954762	J468 E	28.29
952271	J644	0.62
952321	J734	5.06
954721	J750 C	2.1
954722	J750 E	11.38
953371	J808	9.04
953401	J811	17.75
953641	J813	43.77
953651	J815	32.08
953741	J826 C	1.65
953742	J826 E	8.93
953851	J845 C	1.72
953852	J845 E	9.31
953881	J848 C	4.95
953882	J848 E	27.31
954411	J912	14.22
954681	J949	38.89
964791	J955	110.95
965001	J979 C	4.21
965002	J979 E	22.78
965101	J991	41.22
274650	KINCAID ;1U	14.91
274651	KINCAID ;2U	14.93
990901	L-005 E	11.16
290108	LEEDK;1U E	12.43
LTF	LGE-0092018	< 0.01
LTF	MEC	45.67
293516	O-009 E1	6.29
293517	O-009 E2	3.2
293518	O-009 E3	3.52
293715	O-029 E	6.72
293716	O-029 E	3.68

293717	O-029 E	3.39
293771	O-035 E	5.18
LTF	O-066	5.03
293644	O22 E1	4.91
293645	O22 E2	9.53
290021	O50 E	10.79
294763	P-046 E	5.4
LTF	RENSSELAER	0.23
290261	S-027 E	19.38
290265	S-028 E	19.38
295111	SUBLETTE E	1.48
LTF	TRIMBLE	1.3
274853	TWINGROVE;U1	0.75
274854	TWINGROVE;U2	0.75
276150	W2-048 E	2.05
905081	W4-005 C	0.18
905082	W4-005 E	37.17
LTF	WEC	4.14
295109	WESTBROOK E	3.05
909052	X2-022 E	28.53
916211	Z1-072 E	3.92
916221	Z1-073 E	2.94
917502	Z2-087 E	17.34
918052	AA1-018 E	7.93
919221	AA1-146	11.68
919581	AA2-030	11.68
920272	AA2-123 E	1.22
930481	AB1-089	35.4
930741	AB1-122 1O1	35.01
924041	AB2-047 C O1	3.24
924042	AB2-047 E O1	21.68
924261	AB2-070 C O1	3.84
924262	AB2-070 E O1	25.7
925302	AB2-191 E	0.75
925581	AC1-033 C	1.16
925582	AC1-033 E	7.74
925771	AC1-053 C	3.88
925772	AC1-053 E	25.95
927511	AC1-113 1	0.65
927521	AC1-113 2	0.65
926431	AC1-114	1.3
926821	AC1-168 C O1	0.83
926822	AC1-168 E O1	5.54
926841	AC1-171 C O1	1.14
926842	AC1-171 E O1	7.6
927201	AC1-214 C O1	1.66
927202	AC1-214 E O1	5.28

